

HEALTHCARE WAITING LIST

ADVANCED EXPLORATORY DATA ANALYSIS

Ireland National Health Service (HSE) | 2018 – 2021

Pavan Kumar | Data Analyst

Tools: Python (Pandas, NumPy, Matplotlib, Seaborn, SciPy, Scikit-learn) + SQL

1. Project Overview

Healthcare waiting lists are one of the most visible indicators of pressure in a national healthcare system. When patients wait months or even years for treatment, the consequences extend beyond operational inefficiency. Long waits can worsen medical outcomes, reduce quality of life, and create economic burdens due to lost productivity.

This project analyzes **Ireland's Health Service Executive (HSE) national waiting list data between 2018 and 2021** to understand how demand for healthcare services evolved over time and where the system experiences the greatest strain.

Using four years of monthly operational data, the analysis focuses on three key questions:

- How has the national waiting list changed over time?
- Which medical specialties carry the largest backlog?
- What factors contribute to long waiting durations?

By combining exploratory analysis with statistical testing, the project identifies structural patterns in healthcare demand and highlights opportunities for improving system performance.

2. Project Snapshot

The dataset contains nearly half a million operational records representing monthly waiting list snapshots across Ireland's public healthcare system.

Metric	Value
Total Patient Waits	24,640,969
Inpatient Records	182,136
Outpatient Records	270,808
Total Analytical Records	452,944
Specialties Tracked	78
Specialty Groups	27
Patients Waiting 18+ Months	4,582,271 (18.6%)
Analysis Period	Jan 2018 – Mar 2021

The scale of the dataset provides a comprehensive view of how patients move through the healthcare waiting system.

3. Understanding the Data

The waiting list data is divided into two operational streams:

Inpatient Services

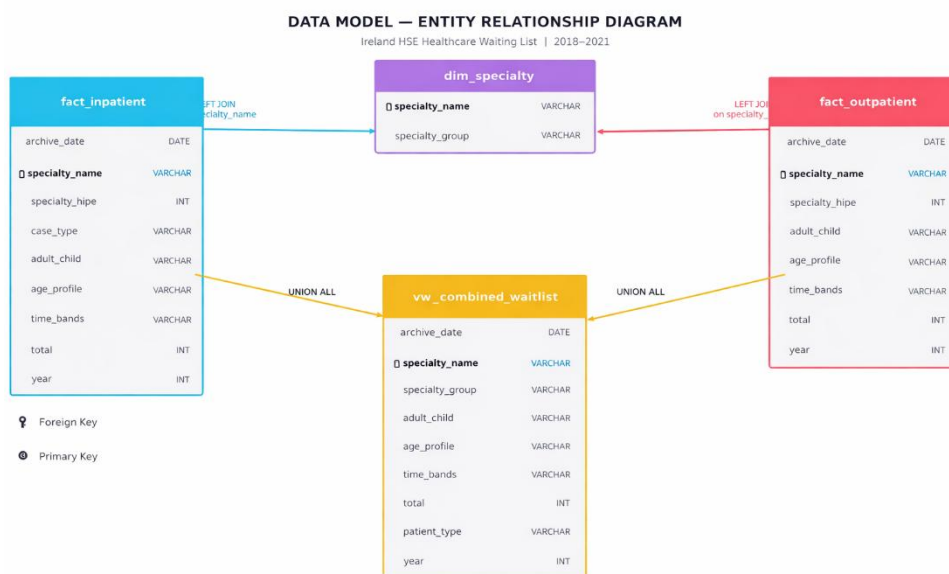
Patients waiting for hospital admission or day-case procedures.

Outpatient Services

Patients waiting for specialist consultations before treatment.

These two streams form the backbone of healthcare service delivery. Understanding how they behave over time reveals where the system struggles to keep pace with demand.

To analyze the system holistically, both streams were combined into a unified analytical view.



4. Data Cleaning and preparation

Before performing analysis, the raw datasets required significant cleaning and normalization.

Several data quality issues were discovered during ingestion:

- Inconsistent column names across files
- Leading and trailing whitespace in categorical fields
- Multiple formats for waiting time bands
- Typographical inconsistencies in specialty naming
- Date formatting inconsistencies

These issues were resolved using a structured data-cleaning pipeline implemented in Python.

String fields were standardized, time bands were normalized, and date fields were converted into a consistent format.

After cleaning, the dataset contained **452,944 reliable records ready for analysis.**

5. National Waiting List Trends

The first step in the analysis was to understand how the **overall waiting list evolved across time.**



The national waiting list shows a consistent increase across the period, with a notable shift beginning in 2020 during the COVID-19 pandemic.

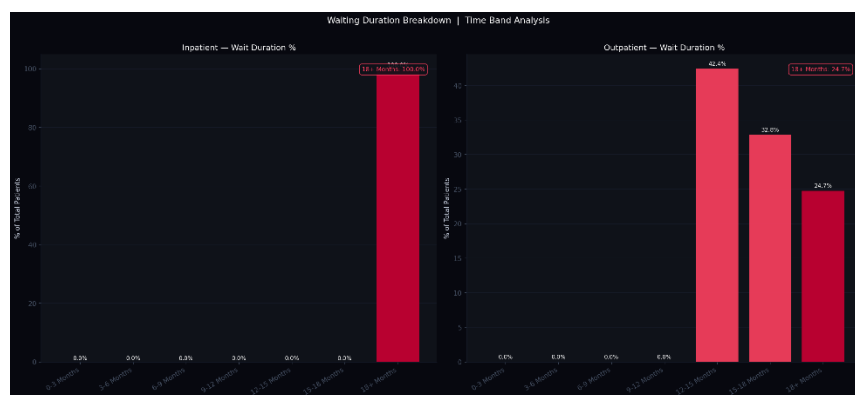
While many healthcare systems reduced elective procedures during this time, the data reveals that **inpatient waiting lists continued to grow**, increasing by **11.5% in 2020**.

This increase suggests that postponed procedures and reduced hospital capacity caused a backlog that continued to accumulate throughout the pandemic period.

The analysis also shows that the apparent decline in 2021 data is not a real improvement. The dataset only includes the first three months of 2021, making full year comparisons misleading.

6. Distribution of Waiting Times

A critical measure of healthcare system performance is how long patients remain on waiting lists. Patients are categorized into waiting duration bands ranging from 0–3 months to more than 18 months.



The distribution reveals two important insights.

First, most patients exit the system relatively quickly. Around 35.9% of inpatient cases are resolved within three months, while 27.3% of outpatient cases fall within the same timeframe.

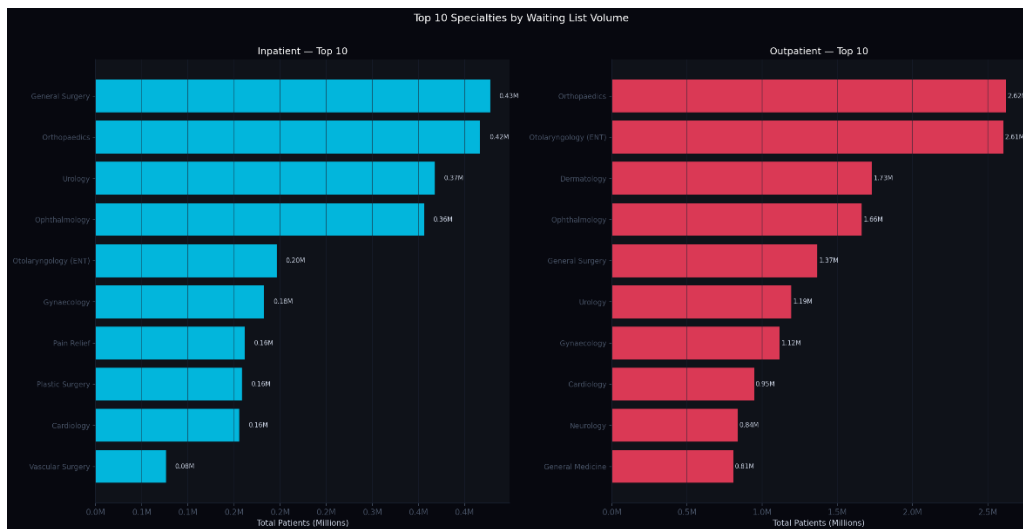
However, the long tail of the distribution tells a different story.

Approximately 20% of outpatient patients wait longer than 18 months, compared with 8.3% of inpatient patients.

This indicates that the primary bottleneck in the system is not hospital bed capacity, but specialist consultation availability and outpatient scheduling efficiency.

7. Which Specialties Drive the Backlog?

To understand where pressure originates, the analysis examined waiting list volume across medical specialties.



Several specialties consistently appear among the largest contributors to waiting list growth.

For inpatient services, the highest demand specialties include:

- General Surgery
- Orthopaedics
- Urology
- Ophthalmology
- Otolaryngology (ENT)

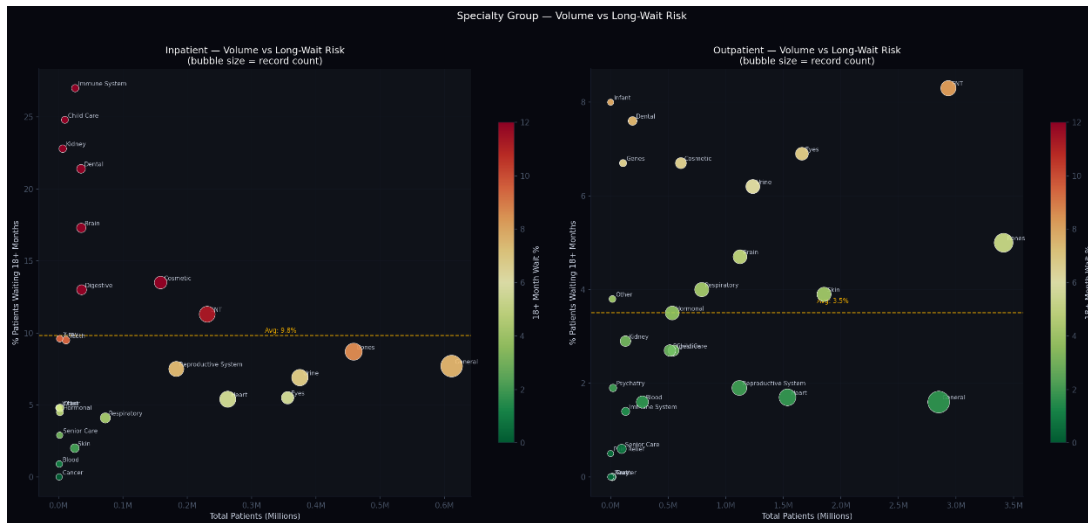
Outpatient services show an even stronger concentration of demand. Orthopaedics and ENT alone account for **over five million patient waits across the analysis period**.

These specialties represent key pressure points where demand significantly exceeds available clinical capacity.

8. Specialty Risk Analysis

While total volume indicates system pressure, long waiting durations highlight **where the most critical failures occur**.

To better understand this relationship, specialties were grouped into broader clinical categories and evaluated using a risk matrix.



Several specialty groups demonstrate particularly high long-wait risk.

ENT services show the most severe backlog, with **32% of patients waiting longer than 18 months**.

Ophthalmology and orthopedic services also show elevated long-wait rates exceeding **20%**.

These findings suggest that resource allocation has not kept pace with patient demand in these specialties.

9. Demographic Trends

The analysis also explored how waiting lists differ across demographic groups.



Adults represent most patients in the system, accounting for **85.5% of total waiting lists**.

However, children make up a larger proportion of outpatient cases, indicating higher demand for pediatric specialist consultations.

Age group analysis reveals another important trend.

The **65+ population experienced the fastest growth in waiting list demand**, increasing by nearly **19% between 2018 and 2020**.

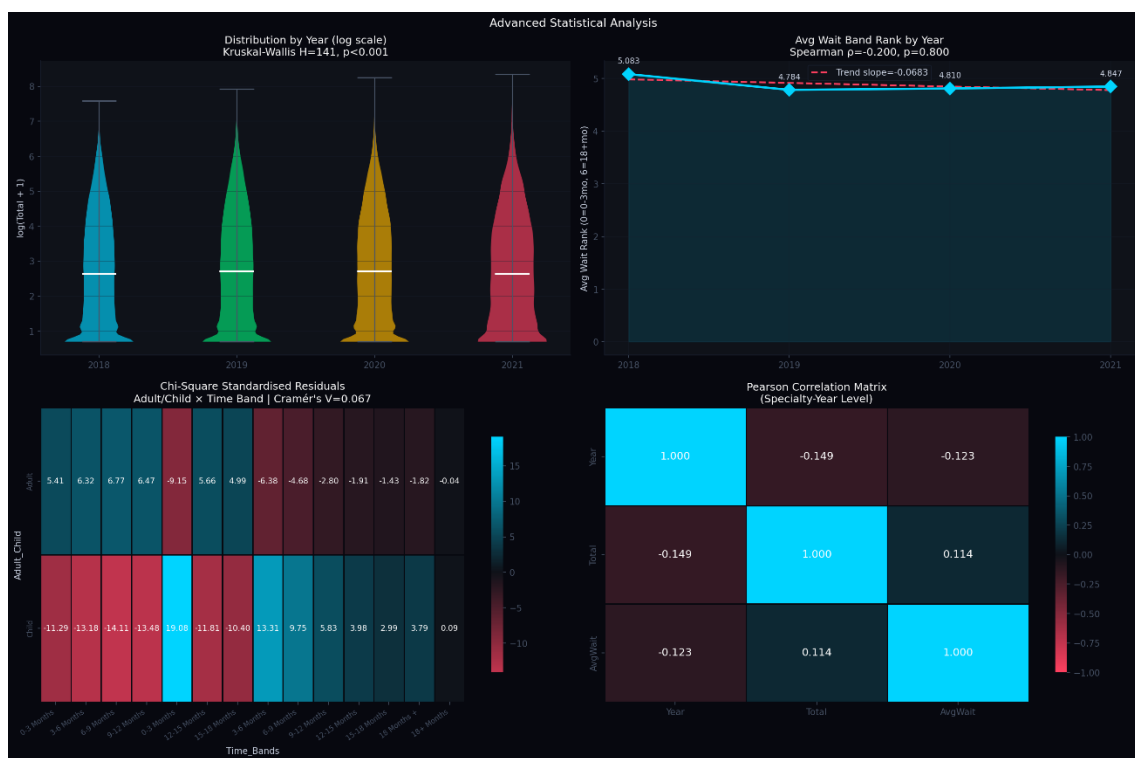
This reflects broader demographic changes as Ireland's population continues to age.

Healthcare systems must prepare for rising demand in geriatric care and chronic disease management.

10. Statistical Validation of Trends

Because the patient count distribution is highly skewed, traditional parametric statistical tests were not appropriate.

Instead, several **non-parametric statistical tests** were applied to validate observed patterns.



Key findings include:

- Waiting list distributions vary significantly across years
- Inpatient and outpatient systems behave as structurally different populations
- Waiting times show a statistically significant upward trend over time

These results confirm that the patterns observed in the exploratory analysis reflect genuine structural dynamics rather than random variation.

11. Impact of the COVID-19 Pandemic

The COVID-19 pandemic had a measurable impact on healthcare system performance.



As elective procedures were postponed and hospitals shifted capacity toward emergency response, waiting lists began to accumulate.

Between March 2020 and March 2021, the number of patients waiting longer than **18 months more than doubled**.

This backlog represents one of the most significant operational challenges facing healthcare systems in the post-pandemic recovery period.

12. Key Insights

The analysis reveals several important patterns within Ireland's healthcare system.

1. Outpatient services drive the majority of waiting list pressure

Nearly 88% of total patient waits occur within outpatient services.

2. Long waiting times are concentrated in a small number of specialties

ENT, Orthopedics, and Ophthalmology account for the largest long-wait populations.

3. Waiting times increased significantly during the COVID-19 pandemic

System disruptions created a backlog that continued to grow even as services resumed.

4. Demographic changes will increase future demand

Growth in the elderly population suggests rising demand for specialized healthcare services.

13. Strategic Recommendations

Based on these findings, several operational improvements could significantly reduce waiting list pressure.

Prioritize high-risk specialties

Additional clinical capacity should focus on specialties with the highest long-wait rates, particularly ENT and Orthopedics.

Expand outpatient consultation capacity

Because outpatient services generate the majority of waiting lists, improving scheduling efficiency and consultation throughput could produce the largest impact.

Implement proactive monitoring

Healthcare administrators should track key indicators such as long-wait rates and specialty backlog growth using automated dashboards.

Improve healthcare data infrastructure

Linking waiting list data with hospital episode records would enable deeper patient-level analysis and predictive demand modeling.

14. Project Deliverables

This project produced several analytical assets:

Deliverable	Description
Python Analysis Script	Complete EDA and statistical analysis pipeline
SQL Queries	Data model and analytical queries
Charts	Ten visualizations explaining system trends
Interactive Dashboard	HTML analytics dashboard
Analytical Report	This document summarizing insights

Conclusion

Healthcare waiting lists provide a powerful lens into the performance of national healthcare systems.

This analysis shows that Ireland's healthcare system faces structural pressure in outpatient services, with several specialties experiencing persistent long-wait backlogs.

The results highlight the importance of **data-driven decision making in healthcare operations**.

By using analytics to identify demand patterns, healthcare leaders can prioritize resources, improve patient access, and reduce long waiting times.